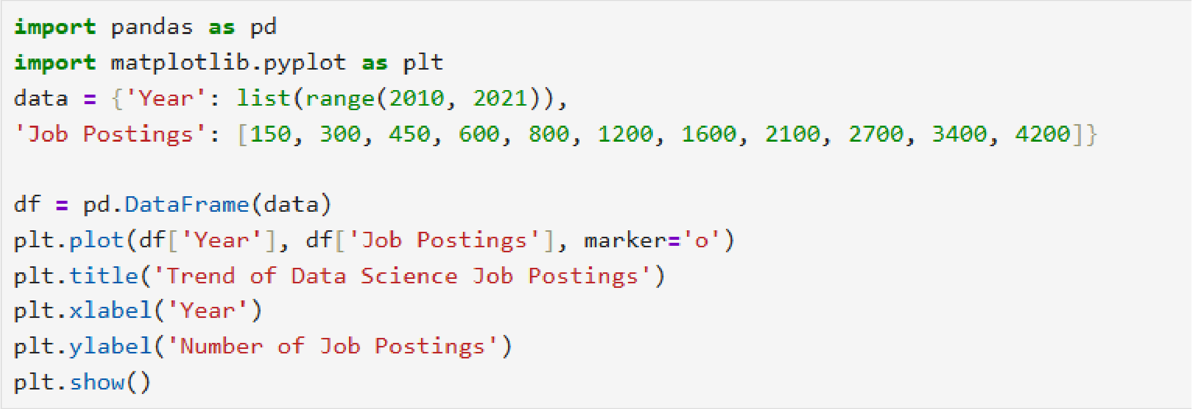
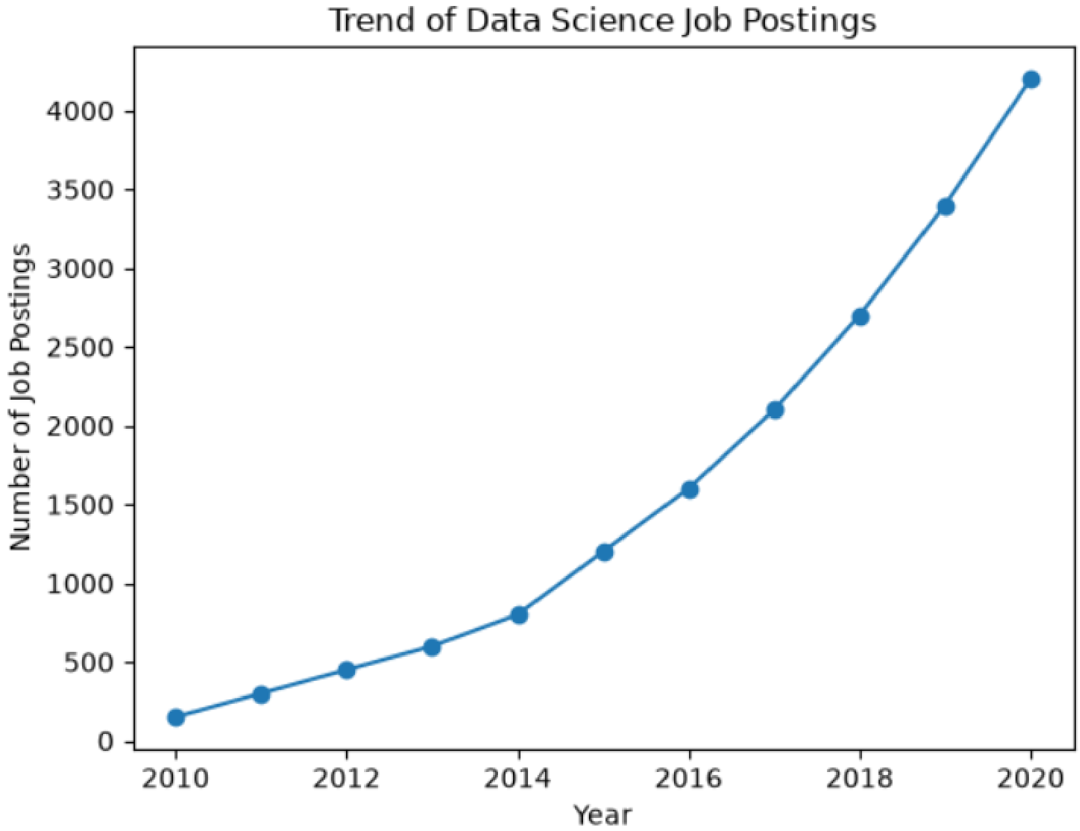
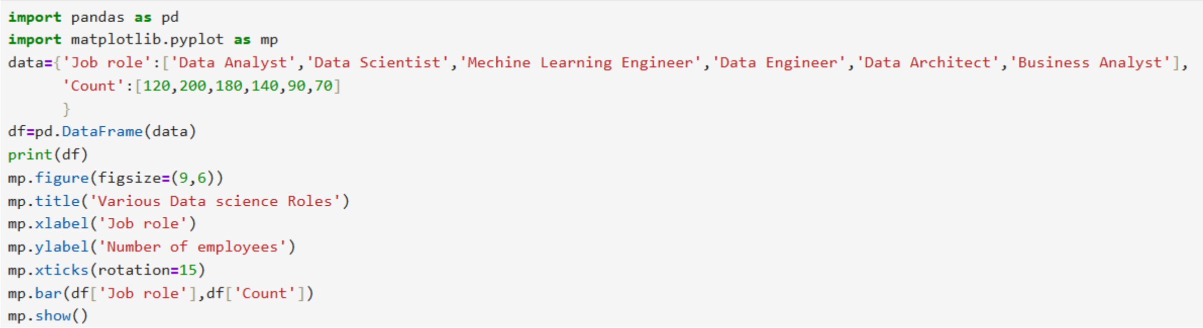


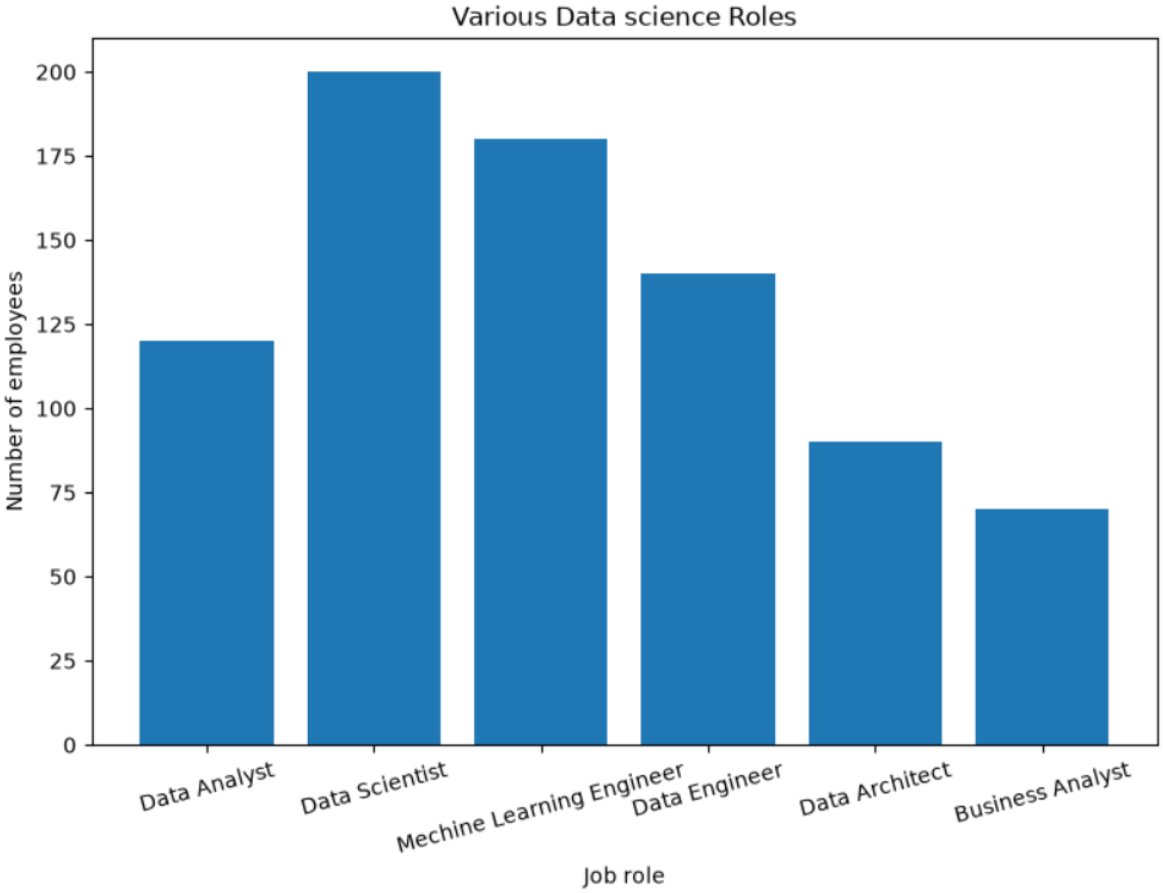
```
import pandas as pd
import matplotlib.pyplot as plt
data = {'Year': list(range(2010, 2021)),
'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}

df = pd.DataFrame(data)
plt.plot(df['Year'], df['Job Postings'], marker='o')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.show()
```





```
import pandas as pd
import matplotlib.pyplot as mp
data={'Job role':['Data Analyst','Data Scientist','Mechine Learning Engineer','Data Engineer','Data Architect','Business Analyst'],
      'Count':[120,200,180,140,90,70]
      }
df=pd.DataFrame(data)
print(df)
mp.figure(figsize=(9,6))
mp.title('Various Data science Roles')
mp.xlabel('Job role')
mp.ylabel('Number of employees')
mp.xticks(rotation=15)
mp.bar(df['Job role'],df['Count'])
mp.show()
```


Various Data science Roles
Number of employees
0
25
50
75
100
125
150
175
200
Data Analyst
Data Scientist
Mechine Learning Engineer
Data Engineer
Data Architect
Business Analyst
Job role

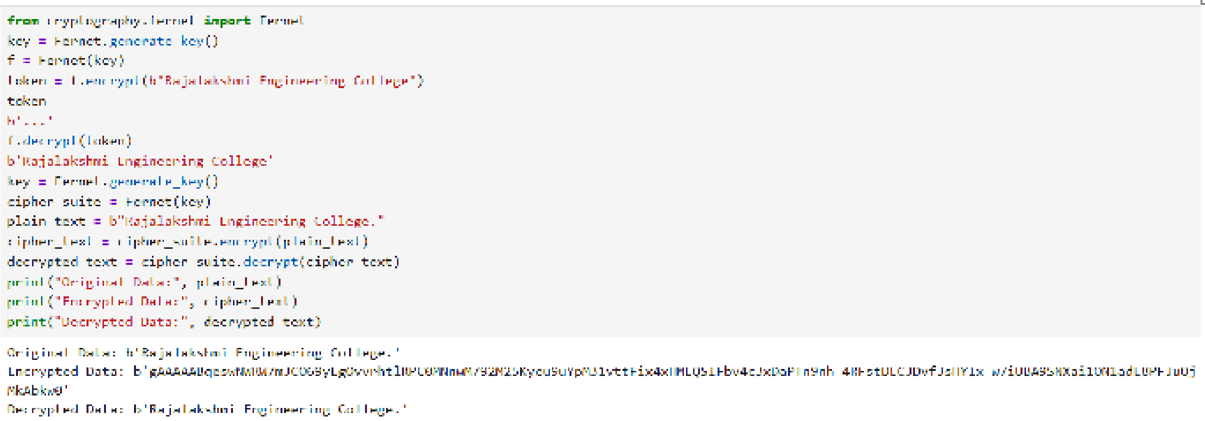
```python
#Structured data
import pandas as pd
data={
    'Name':['Yamuna','Yuga','Yogitha','Merylin'],
    'Age':[20,21,18,19],
    'ID':[201,208,209,210]
}
df=pd.DataFrame(data)
print("Structured data")
print(df)

#Semi Structured data
import json
data={
    'Name':'Bob',
    'Age':20,
    'Id':201,
    'Address':'Chennai'
}
print("Semi Structured Data")
print(json.dumps(data,indent=5))

#Unstructured data
text="""
JSON stands for JavaScript Object Notation.
It organizes data using key-value pairs,
but it is not arranged in rows and columns like a table.
"""
print("Unstructured data")
print(text)
```

```
Structured data
      Name  Age   ID
0   Yamuna   20  201
1     Yuga   21  208
2  Yogitha   18  209
3  Merylin   19  210
Semi Structured Data
{
     "Name": "Bob",
     "Age": 20,
     "Id": 201,
     "Address": "Chennai"
}
Unstructured data

JSON stands for JavaScript Object Notation.
It organizes data using key-value pairs,
but it is not arranged in rows and columns like a table.
```



```python
from cryptography.fernet import Fernet
key = Fernet.generate key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
b'...'
f.decrypt(token)
b'Rajalakshmi Engineering College'
key = Fernet.generate_key()
cipher suite = Fernet(key)
plain text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)
decrypted text = cipher suite.decrypt(cipher text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted text)
```

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data: b'gAAAAABqeswNM0W/mJC069yLgOvvrhtl0PL0MNmwM/92M25KyeuSuYpMJ1vttFix4xIMLQS1Fbv4cJxDsFIn9nh 4RFstULCJDvfJsHY1x w/iUUA9SNXai1UN1adL0PFJuOj
MkAbkw0'
Decrypted Data: b'Rajalakshmi Engineering College.'
```